

Experiments

What matters for routing?

Stephanus Park, 22.04.2026

This talk focuses on what actually matters for routing: not similarity, but utility.

I'll show why our current signals fail and how that leads to a new direction.

Agenda

What matters for routing?

- Problem
 - Initial hypothesis
 - Evaluation protocol
- Results
 - Core finding
 - Failure diagnosis
 - Pivot
- Takeaways

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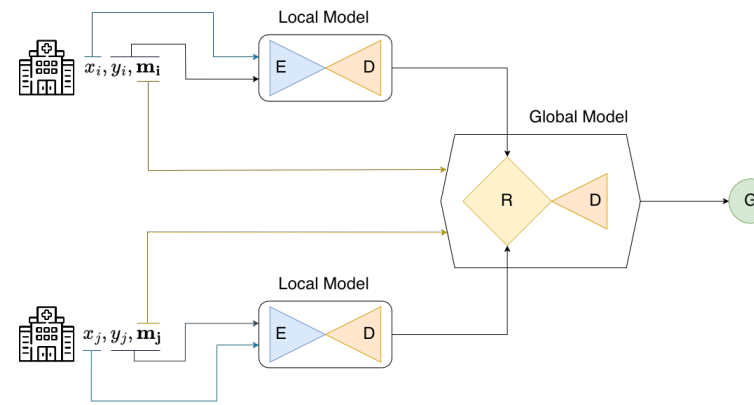
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I'll follow a causal pipeline:

- Define the problem
- Test a hypothesis
- Evaluate
- Diagnose failure
- Derive a pivot

Introduction

What am I actually doing



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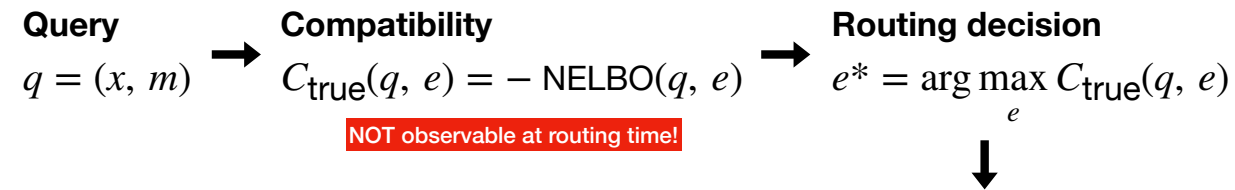
Fig. 1: Overview of a modular generative framework enabling privacy-preserving, domain-adapted, and efficient synthetic data generation across multiple institutions.

Problem analysis

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Problem analysis

Routing is a compatibility estimation problem



A CVAE is trained to maximize: $\log p_\theta(x|m)$, but this is intractable, so we optimize its variational bound:

$$\text{NELBO}(q, e) = \underbrace{\text{reconstruction error}}_{\text{fit to data}} + \underbrace{\text{latent regularization}}_{\text{distribution mismatch}}$$

Utility

- Actual generative performance
- Measured again by NELBO

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(Sohn et al., 2015)

The problem is selecting the best expert for a given query.

Formally, routing is a compatibility estimation problem:

Compatibility is defined as expected utility — negative NELBO.

[CLICK]

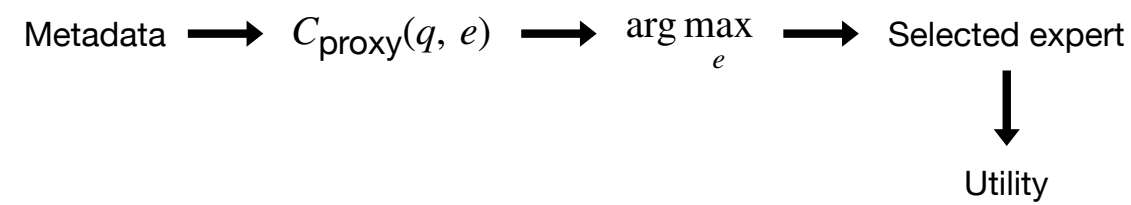
The key difficulty is that this quantity is **not observable at routing time**.

Initial hypothesis

Metadata similarity $\approx$ compatibility

Assume: $C_{\text{proxy}} \approx C_{\text{true}}$

C_{proxy} defined via metadata-based similarity (domain distance)



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(Kömen et al., 2024)

Since true compatibility is not observable, we approximate it using metadata similarity.

Prior work shows that batch effects persist even in strong models, implying that domain metadata captures meaningful variation.

This motivates using metadata similarity as a proxy for compatibility. [PAUSE]

Metadata similarity

Proxy definition

BreakHis

$$s(q, e) = \exp\left(-\frac{|m_q - m_e|}{\tau}\right)$$

Camelyon17

$$s(q, e) = \mathbf{1}[q = e]$$

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We define a simple proxy similarity based on the structure of the metadata.

[CLICK]

For BreakHis, magnification is ordinal, so we use a smooth distance-based similarity.

[CLICK]

For Camelyon17, site is categorical, so we use exact matching.

Bridge:

We use a minimal and faithful encoding of metadata.

This is important: the goal is not to build the best model,
but to test whether the signal exists at all.

Evaluation protocol

Improvement requires consistency across all three gates

E1: Utility	NELBO (↓ better)	
E2: Ranking quality	Top-1 accuracy	Spearman correlation
E3: System outcome	Oracle gap (%)	

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We test whether proxy signals can recover true compatibility.

We evaluate this using three gates.

E1 defines the ground-truth compatibility via NELBO.

In the experiments, we do not optimize E1 directly.

Instead we test:

- Whether proxy signals can recover its ranking (E2), and
- What happens when they fail (E3)

Results

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Now I'll show the key empirical results.

Result 1 — Proxy-based compatibility fails

Metadata is a weak proxy for compatibility

BH = BreakHis, C17 = Camelyon17

Method	Top-1 (BH / C17)	Spearman (BH / C17)	Gap (BH / C17)	Stable?	Pass
Metadata (baseline)	0.33 / 0.27	0.16 / 0.10	0.008 / 0.001	Yes	No
Learned methods (best non-oracle)	0.28 / 0.31	0.14 / 0.15	worse than baseline	No	No
Oracle ranking*	perfect	perfect	0	No	No

* uses true NELBO (not deployable)

Tab. 1: Empirical evidence that proxy-based compatibility (metadata, learned predictors) is not utility-aligned (–NELBO), motivating compatibility as expected utility for expert routing.

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This table shows compatibility estimation results across BreakHis (BH) and Camelyon17 (C17).

Proxy-based methods fail to recover the correct expert ranking (low Top-1, weak Spearman), despite small oracle gap indicating weak utility separability.

Small oracle gap does not imply good compatibility estimation —
it only indicates that the utility landscape is shallow. Correct expert identification still fails.

Even with simple and expressive models, we cannot recover a consistent ranking signal.

This suggests that compatibility is not recoverable from the observables.

Result 2 — Enforcing metadata usage fails

Conditioning does not translate metadata into utility gains

Setup: Metadata injected into encoder and decoder of CVAE

Metric	BreakHis	Camelyon17	Interpretation
E1: Utility	Worse than baseline	Worse than baseline	Does not improve utility despite metadata signal
E2: Ranking quality	Top-1 ↑, Spearman ↓	Top-1 ↑, Spearman ↓	Fails to recover NELBO ranking
E3: System outcome	No gap reduction	No gap reduction	No improvement in expert selection outcome

Tab. 2: Enforcing metadata usage via CVAE conditioning does not improve utility (–NELBO), ranking quality, or system outcome across domains.

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We now test whether forcing metadata usage helps.

We inject metadata directly into the model.

Metadata conditioning fails to translate metadata signal into utility-aligned compatibility—no gains in NELBO, ranking, or expert selection outcome.

So the issue is not weak usage — it’s misalignment.

Core findings

Bottleneck: compatibility estimation (utility-aligned)

Observation	Implication
Oracle routing achieves high Top-1	Correct expert exists and is identifiable
Proxy signals (metadata / conditioning) change ranking	Compatibility signal exists
Utility (NELBO) does not improve	Signal is not aligned with utility

Tab. 3: Compatibility ≠ similarity. Only utility (–NELBO) defines correct expert selection.

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From these results, we can isolate the bottleneck.

The correct expert exists (oracle works).
Proxy signal affect ranking (signal exists).

But utility does not improve.

Therefore, the proxy signal is **not aligned with utility**.

The bottleneck is compatibility estimation — not routing.

Failure diagnosis

Where the pipeline breaks

Stage	Status
Expert quality	✓
Routing mechanism	✓
Compatibility estimation	✗

Tab. 4: Failure diagnosis—pipeline breakdown is localized to compatibility estimation, while expert quality and routing remain functional.

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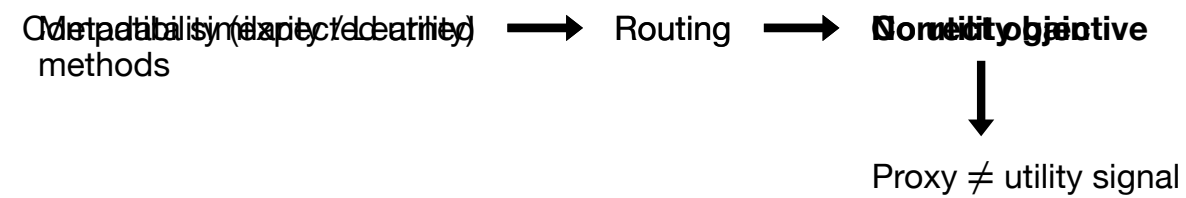
This just localizes the previous finding in the pipeline.

Experts are fine.
Routing changes behavior.

Compatibility estimation is where the system breaks.

Pivot

Proxy compatibility is fundamentally misaligned



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These results indicate a structural failure:

Proxy signals such as metadata or learned predictors do not align with true utility.

This suggests that compatibility cannot be treated as similarity.

[CLICK]

Instead, it must be defined as expected utility, measured by NELBO.

Pivot

Compatibility must be learned from utility

- Not fixable by:
 - Richer metadata → already forced (Result 2)
 - More expressive models → probes failed (Result 1)
- Therefore:
 - Observables do not encode utility-aligned signal

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The misalignment of the proxy signal cannot be fixed by:

- “Richer metadata (already tested)”
- “More expressive models (probes failed)”

Therefore, the signal is not in the observables.

If compatibility cannot be recovered from observables,
It must be learned from utility.

Bridge:

How can we achieve this?

Pivot

Reformulation — Compatibility as supervised ranking

Query	$q = (x, m)$
Target	$C_{\text{true}}(q, e) = -\text{NELBO}(q, e)$
Task	Learn ranking over experts per query
Objective	Optimize ordering, not exact values

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We reformulate compatibility estimation as a supervised ranking problem.

The target is expected utility (- NELBO).

The goal is not to predict exact values,
But to recover the correct expert ranking.

Takeaways

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Finally, the main takeaways.

Takeaways

What is actually interesting

- Metadata is **not sufficient** for expert routing
- Compatibility must be defined as **expected utility**
- Routing fails because **proxy $\neq$ objective**
- Next objective:
 - Learn compatibility directly

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The key takeaway is:

Metadata is not sufficient for routing.

Compatibility must be defined as expected utility.

The failure arises from a misaligned proxy objective.

The next step is to learn compatibility directly.

Thank you!

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References

- Kömen, J., Marienwald, H., Dippel, J., & Hense, J. (2024). Do Histopathological Foundation Models Eliminate Batch Effects? A Comparative Study (Version 1). arXiv. <https://doi.org/10.48550/ARXIV.2411.05489>
- Sohn, K., Lee, H., & Yan, X. (2015). Learning structured output representation using deep conditional generative models. In C. Cortes, N. Lawrence, D. Lee, M. Sugiyama, & R. Garnett (Eds.), *Advances in neural information processing systems* (Vol. 28). Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2015/file/8d55a249e6baa5c06772297520da2051-Paper.pdf

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Appendix

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Appendix: Practical computation

How NELBO is computed

For each expert e :

1. Encode: $z \sim q_{\phi_e}(z \mid x, m)$
2. Reconstruct: $\log p_{\theta_e}(x \mid z, m)$
3. KL term: $KL(q_{\phi_e}(z \mid x, m) \parallel p_{\theta_e}(z \mid m))$
4. Combine: NELBO = reconstruction loss + KL

Appendix: Compatibility estimators

Compatibility failure is independent of estimator complexity

Method	Signal used	Capacity	Result
Constant mean	None	None	Fails
Metadata prior	Metadata	Low	Fails
Linear regression	Features + metadata	Linear	Fails
MLP regression	Features + metadata	Nonlinear	Fails
Utility probe	Normalized features	Structured	Fails
Semi-oracle	Partial utility stats	High	Unstable
Oracle	True NELBO	Perfect	No system gain

Tab. A1: Comparison of compatibility estimators across signal types and model capacities. All proxy-based estimators fail to recover utility-aligned compatibility (–NELBO), regardless of complexity, while oracle access confirms recoverability without yielding system gains.

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